

Institute of Business Administration

Course: Comp. Communications & Networking	Exam: Final
Instructors: Waseem Arain / Farhan Ahmed Siddiqui	Time: 150 Minutes
Class: BSCS	Date: 17 th May, 2022

Name:

ERP No:

PART B - SUBJECTIVE (80 Points) (Time 150 Minutes)

Short Question/Answer Section: (Attempt any 10 each has 2 points) 20 points

1.

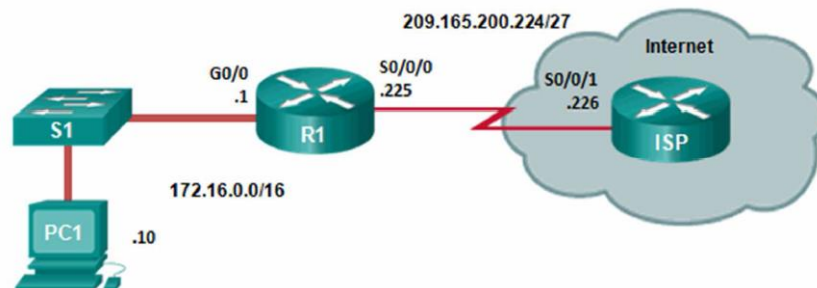
```
R1# show ip nat translations
```

```
Pro Inside global      Inside local    Outside local    Outside global
tcp 209.165.200.225:1405 10.6.15.2:1405 209.165.202.141:80 209.165.202.141:80
tcp 209.165.200.225:1406 10.6.15.1:1406 198.51.100.3:80    198.51.100.3:80
```

Refer to the exhibit. Based on the output that is shown above, what type of NAT has been implemented?

2. In an IP Header, a packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?

3. R1 was configured with the static route command `ip route 209.165.200.224 255.255.255.224 S0/0/0` and consequently users on network 172.16.0.0/16 are unable to reach resources on the Internet. How should this static route be changed to allow user traffic from the LAN to reach the Internet?



4. Assume a TCP client expects to receive byte 2001, but it receives a segment with sequence number 1201. What is the reaction of the TCP client to this event? Can you justify the reaction?

5. Suppose that N switches supporting K VLAN groups are to be connected via a trunking protocol. How many ports are needed to connect the switches? Justify your answer

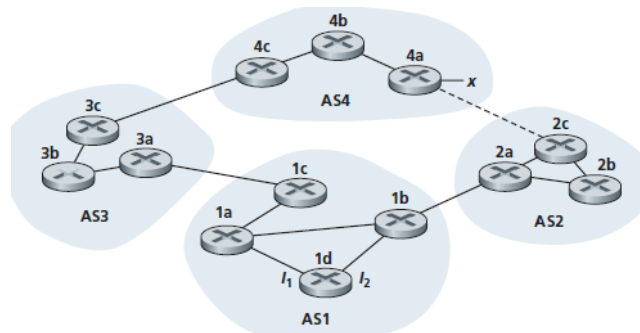
6. Discuss why each input port in a high-speed router stores a shadow copy of the forwarding table.

7. What is HOL blocking? Does it occur in input ports or output ports?
8. What field in the IP header can be used to ensure that a packet is forwarded through no more than N routers?
9. Suppose there are three routers between a source host and a destination host. Ignoring fragmentation, an IP datagram sent from the source host to the destination host will travel over how many interfaces? How many forwarding tables will be indexed to move the datagram from the source to the destination?
10. Suppose an application generates chunks of 40 bytes of data every 20 msec, and each chunk gets encapsulated in a TCP segment and then an IP datagram. What percentage of each datagram will be overhead, and what percentage will be application data?
11. Compare and contrast the IPv4 and the IPv6 header fields. Do they have any fields in common?
12. What is meant by an area in an OSPF autonomous system? Why was the concept of an area introduced?

Detailed Question/Answers Section (Attempt All, choose between either/or questions) 60 points

1. Consider the network shown below. Suppose AS3 and AS2 are running OSPF for their intra-AS routing protocol. Suppose AS1 and AS4 are running RIP for their intra-AS routing protocol. Suppose eBGP and iBGP are used for the inter-AS routing protocol. Initially suppose there is *no* physical link between AS2 and AS4. (4)

- a. Router 3c learns about prefix x from which routing protocol? →
- b. Router 3a learns about x from which routing protocol? →
- c. Router 1c learns about x from which routing protocol? →
- d. Router 1d learns about x from which routing protocol? →



OR

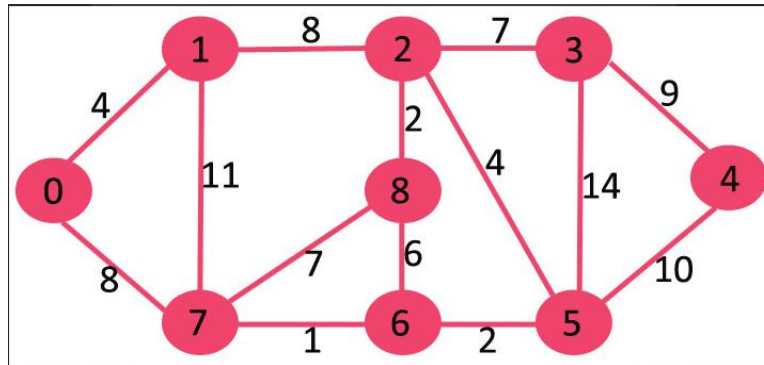
1. Show the functioning of IEEE802.11-MAC (WiFi) through a timing diagram showing its features like RTS/CTS/ACK/ NAV using sender node, receiving node and hidden node, with respect to sender node, by denoting them with S, R and H respectively. (4)

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-
-
2. Test the code word 010101100011 are correct, assuming they were created using an even parity Hamming Code. If it is incorrect, indicate what the correct code word should have been. Also, indicate what the original data was. (6)

OR

2. How does public-key system work to provide: (write only through notation, using **A** for Alice and **B** for Bob, **K** for key + for public and - for private) (6)
- a. confidentiality b. authentication

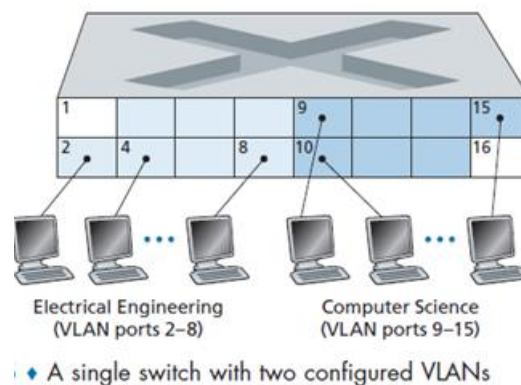
3. Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from 7 to all network nodes. Show how the algorithm works by computing a complete table. (5)



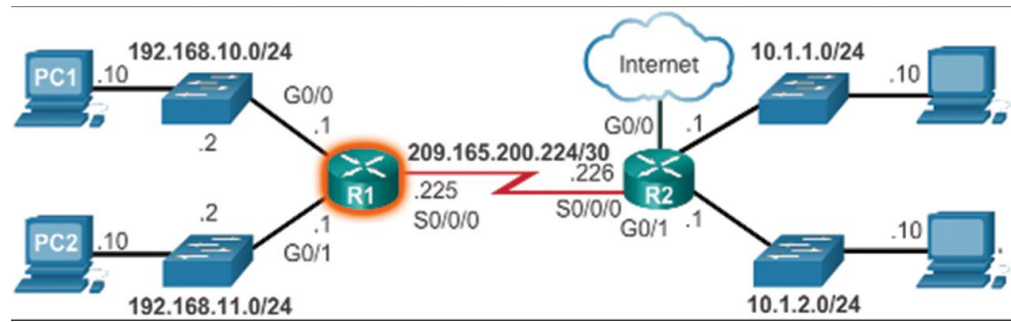
4. Consider sending a 2400-byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? (4)

OR

4. Consider the single switch VLAN in Figure below and assume an external router is connected to switch port 1. Assign IP addresses to the EE and CS hosts and router interface. Trace the steps taken at both the network layer and the link layer to transfer an IP datagram from an EE host to a CS host (4)



5. Make a routing table for Router R1 & R2 using the configuration given in the figure. (6)



Complete Routing table for router R1

Mask	Network Address	Next Hop	Interface

Complete Routing table for router R2

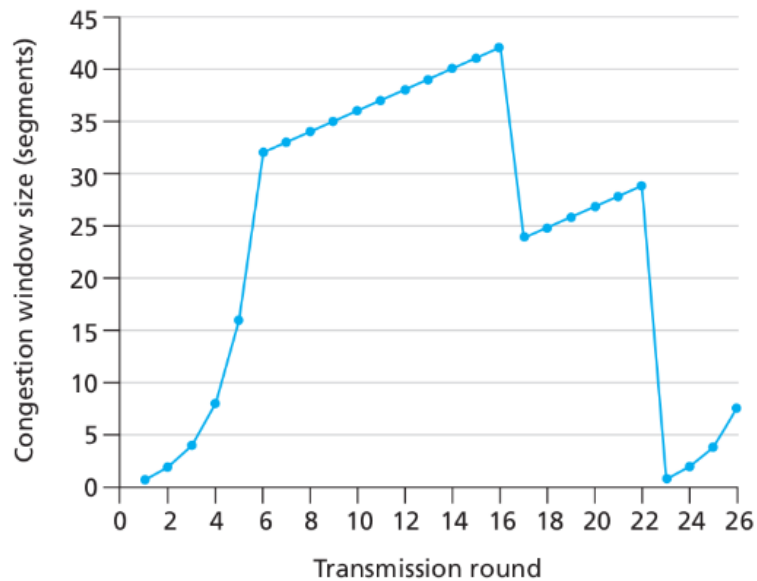
Mask	Network Address	Next Hop	Interface

6. We saw that TCP waits until it has received three duplicate ACKs before performing a fast retransmit. Why do you think the TCP designers chose not to perform a fast retransmit after the first duplicate ACK for a segment is received? (4)

OR

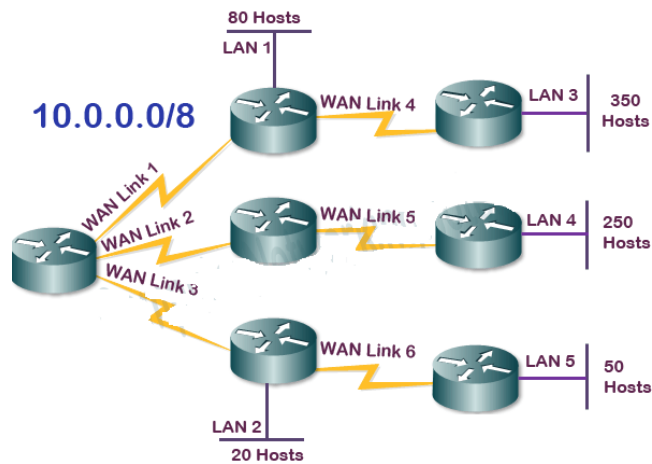
6. Consider the 5-bit generator, $G=10011$, and suppose that D has the value 1010101010. What is the value of R ? (4)

7. Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. In all cases, you should provide a short discussion justifying your answer. (12)



- Identify the intervals of time when TCP slow start is operating.
- Identify the intervals of time when TCP congestion avoidance is operating.
- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- What is the initial value of ssthresh at the first transmission round?
- What is the value of ssthresh at the 18th transmission round?
- What is the value of ssthresh at the 24th transmission round?
- Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of ssthresh?
- Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the ssthresh and the congestion window size at the 19th round?

8. You are given the network address 10.0.0.0/8 to develop an address scheme for the network shown in the Topology diagram. All WAN links required two IP addresses. VLSM will be used so that the addressing requirements can be met. (10)



Subnet Table:

Subnet Description	Number of hosts needed	Network Address/ CIDR	First Usable IP	Broadcast Address

9. Consider a datagram network using 32-bit host addresses. Suppose a router has four links, numbered 0 through 3, and packets are to be forwarded to the link interfaces as follows: (4)

Destination Address Range	Link Interface
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1
11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
otherwise	3

- a. Provide a forwarding table that has five entries, uses longest prefix matching, and forwards packets to the correct link interfaces.

- b. Describe how your forwarding table determines the appropriate link interface for datagrams with destination addresses:

Destination Addresses	Interface
11001000 10010001 01010001 01010101	
11100001 01000000 11000011 00111100	
11100001 10000000 00010001 01110111	

10. Consider the network shown below, and assume that each node initially knows the costs to each of its neighbors. Consider the distance-vector algorithm and show the distance table entries at node z. (5)

